House sales in King County, USA: analysis and prediction on the prices

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Abstract

This study investigates the problem of predicting residential property prices using structural and geographic characteristics. The analysis is conducted on the House Sales in King County dataset, which includes more than 21,000 housing transactions. Several machine learning models are compared, including Linear Regression, Random Forest, Multilayer Perceptron and Gradient Boosted Trees, using both Holdout and Cross Validation strategies. The results show that ensemble-based models achieve the highest predictive accuracy and generalisation performance. Feature selection further highlights the importance of property size, location and quality, while reducing model complexity without degrading performance.

Keywords

Machine Learning — House Price Prediction — Ensemble Methods

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Introduction

The real estate market is one of the most significant and complex economic sectors, as the selling price of a home emerges from the interaction of multiple structural, geographic, and qualitative factors. Understanding which characteristics most influence a property's value and accurately estimating its price is of great interest, not only for real estate agents and investors, but also for potential buyers wishing to assess the appropriateness of a price given the features offered.

This study aims to address two fundamental research questions: Is it possible to predict a home's price based on its structural and geographic characteristics? If so, what are the characteristics that most influence a home's selling price?

To answer these questions, the House Sales in King County, USA dataset, available on the Kaggle platform [1], was chosen. The dataset contains 21,614 observations describing home sales in King County, Washington, between May 2014 and May 2015, and includes 21 variables describing the main characteristics of the properties, including size, number of rooms, condition, geographic location, and sale price. The target variable of the analysis is the

sale price, while the remaining variables represent the features used for prediction.

The objective of the analysis is twofold. First, it aims to identify the best-performing machine learning model and training technique for predicting house prices. Second, it seeks to determine which characteristics have the greatest impact on the target variable.

This report is organised as follows:

- Data exploration: the main characteristics of the dataset and the distribution of the price variable are analysed.
- Data cleaning and preprocessing: the data is made suitable for model training, for example, by excluding some variables and transforming others.
- Models: Four different machine learning models are described and compared, using different training and validation techniques.
- Evaluation: The models are evaluated and compared based on quantitative metrics to identify the most accurate solution.
- Conclusions: The results are summarised, highlighting the most influential features and discussing the limitations of the analysis.

1. Data Exploration

The data exploration phase aims to provide an initial understanding of the dataset structure and the variables involved in the analysis, with particular attention to the distribution of the target variable. Since the goal of the analysis is to predict housing prices, the dependent variable of the study is price, a continuous numerical variable on a ratio scale representing the selling price of the property expressed in US dollars.

The dataset includes a set of explanatory variables describing the main structural, qualitative, and geographical characteristics of the properties. The variables available in the dataset are briefly described below:

- id (numeric – nominal): unique identifier of the property
- date (temporal): date of sale of the house
- price (numeric – ratio): sale price of the house (target variable)
- bedrooms (numeric – ratio): number of bedrooms
- bathrooms (numeric – ratio): number of bathrooms, possibly expressed as fractional values
- sqft_living (numeric – ratio): living area in square feet
- sqft_lot (numeric – ratio): total lot area in square feet
- floors (numeric – ratio): number of floors in the house
- waterfront (binary): indicator of the presence (1) or absence (0) of a waterfront view
- view (numeric – ordinal): discrete index measuring view quality
- condition (numeric – ordinal): overall state of preservation of the property
- grade (numeric – ordinal): overall rating of the house based on construction quality and finishes
- sqft_above (numeric – ratio): living area above ground level
- sqft_basement (numeric – ratio): basement area
- yr_built (numeric – interval): year the property was built
- yr_renovated (numeric – interval): year of last renovation (0 if never renovated)
- zipcode (categorical – nominal): ZIP code of the area where the house is located
- lat (numeric – ratio): geographic latitude
- long (numeric – ratio): geographic longitude
- sqft_living15 (numeric – ratio): average living area of the 15 closest neighboring houses
- sqft_lot15 (numeric – ratio): average lot size of the 15 closest neighboring houses

These variables can be grouped into three main categories: structural features, qualitative attributes, and location-related features. The presence of ordinal, binary, and categorical variables motivates the need for an appropriate preprocessing phase, particularly with respect to feature encoding and normalisation, before applying machine learning models.

A preliminary inspection of descriptive statistics highlights a high variability in housing prices. The minimum observed price is 75,000 USD, while the maximum reaches 7,700,000 USD. The mean selling price is approximately 540,529 USD, whereas the median is substantially lower (367,689 USD), indicating a strongly right-skewed distribution influenced by a relatively small number of high-end properties.

The distribution of the price variable, shown in Figure 1, confirms this behaviour, displaying a pronounced positive skewness with a long right tail. Such a distribution makes the target variable poorly suited for direct use in regression-based models and distance-based algorithms.

For this reason, a logarithmic transformation of the selling price was applied. The resulting variable, log_price, exhibits a considerably more symmetric

distribution, as illustrated in Figure 2, and is therefore more suitable for subsequent modeling.

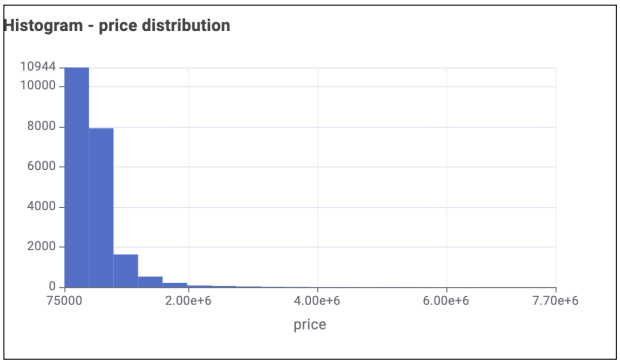


Figure 1: Histogram of price distribution

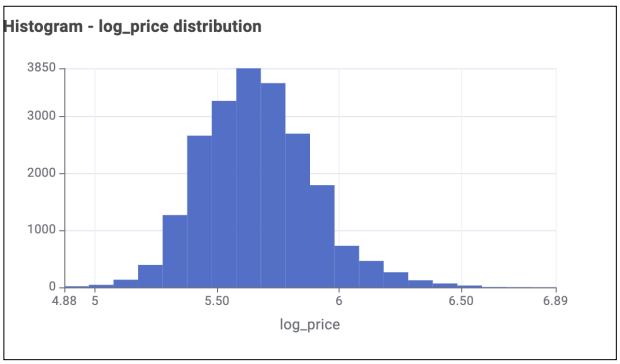


Figure 2: Histogram of log_price distribution

The analysis of the correlation structure reveals that the selling price is strongly associated with the living area (sqft_living, correlation equal to 0.70) and with the construction grade of the property (grade, 0.67). Moderate correlations are also observed with the number of bathrooms and other surface-related variables, while features such as lot size and overall condition show a weaker relationship with the target variable. In addition, the presence of high correlations among multiple surface-related variables provides early evidence of multicollinearity, particularly among different measures of property size, with relevant implications for model stability and anticipating the outcomes of the subsequent feature selection phase. Overall, the data exploration provides a concise yet informative overview of the dataset and supports the methodological choices adopted in the subsequent stages of preprocessing and modeling.

2. Preprocessing

Data preprocessing represents a crucial step in the development of reliable and accurate machine learning models, as the quality of the input data directly affects model performance. In this phase, several operations were carried out to improve the consistency, interpretability and robustness of the dataset before model training.

2.1 Data Cleaning

As a first step, duplicate observations were identified and removed from the dataset. Approximately 200 duplicated records were found and discarded to prevent biased learning, which could otherwise lead the models to overemphasize specific properties appearing multiple times in the data.

The dataset was subsequently inspected for missing values. No missing observations were detected, and therefore no imputation procedures were required.

2.2 Target Variable Transformation

During the data exploration phase, the target variable price was shown to exhibit strong positive skewness, characterised by a long right tail. Such a distribution can negatively affect the performance of regression-based and distance-based models, as it violates the assumption of normally distributed residuals and increases sensitivity to extreme values.

To address this issue, the feature price was log-transformed [2] . This transformation helps stabilise variance, reduces the influence of extreme values and allows the models to better capture the underlying relationships between the predictors and the target variable. The transformed variable, referred to as log_price, was used as the dependent variable in all subsequent analyses.

2.3 Feature Transformation

The variable yr_renovated originally encoded the year of the last renovation or a value of zero if the property had never been renovated. To simplify interpretation and focus on the presence or absence of renovation rather than its timing, this feature was converted into a binary indicator. The transformation enables a clearer distinction between renovated and non-renovated properties, simplifying interpretation while maintaining the most relevant information.

Subsequently, count variables such as the number of bedrooms and bathrooms were preserved in their original numeric form, since they represent quantitative information. However, they were normalized using z-score standardization [3] along with other numeric features to reduce scale differences across variables. This step was particularly important given the wide ranges observed in variables such as sqft_living (from 290 to 13,540) and sqft_lot (up to 1,650,839), and it contributes to improved model stability.

2.4 Feature Selection and Removal

Finally, several features were excluded from the analysis due to their limited predictive value. The id variable was removed because it uniquely identifies each record (house) does not carry meaningful information for price prediction. Similarly, zipcode was excluded because it represents a nominal geographic identifier without intrinsic ordinal meaning and would require high-dimensional encoding, while largely spatial information is already captured by latitude and longitude. Finally, the date variable was also excluded because it refers to the date of sale, which is determined after the price has been set and therefore can't be considered a value influencing the price.

Overall, these preprocessing steps were designed to enhance data quality and support an efficient model training process, ultimately leading to more stable and accurate predictive performances.

3. Models

The following section presents the machine learning models selected for predicting the price variable and the associated KNIME nodes:

Linear Regression - LR: the most well-known class of estimation models, it identifies a functional relationship between a target variable and a set of explanatory attributes. The target variable is predicted as a linear combination of the independent variables, with weights chosen to minimize the error. In Knime, it is implemented through the LinearRegression (3.7) and Weka Predictor (3.7) nodes, using the Akaike criterion for model selection [4]

- Random Forest - RF: based on bagging (Bootstrap Aggregating), which is an ensemble learning algorithm that generates multiple random subsets of the dataset, trains a model on each subset, and combines their predictions to produce the final output. Random Forest implements bagging using decision trees as learners [5]. In KNIME, it is implemented through the Random Forest Learner (Regression) and Random Forest Predictor (Regression) nodes.
- Multilayer Perceptron - MP: uses artificial neural networks organized in multiple layers of neurons. Each neuron receives inputs from all neurons in the previous layer, processes them through an activation function, and transmits the result to the neurons in the next layer. Training is performed using back-propagation, which relies on gradient descent to minimize the error [6]. In KNIME, it is implemented through the MultilayerPerceptron (3.7) and Weka Predictor (3.7) nodes.
- Gradient Boosted Trees - GBT: a type of boosting method, which is an ensemble learning approach that builds a predictive model by sequentially combining weak models to reduce previous errors [7]. Error correction is driven by gradient descent on a loss function. In KNIME, it is implemented using the Gradient Boosted Trees Learner and Gradient Boosted Trees Predictor nodes, following Friedman's approach [8].

These models were selected to compare linear and non-linear approaches, as well as single-model and ensemble-based techniques, and to assess their ability to capture complex relationships in the data.

All models were trained using KNIME default parameters.

3.1 Holdout

The Holdout strategy was implemented by randomly splitting the dataset into two parts, using a fixed random seed to ensure reproducibility of the experiment: 70% of the records were used to train the models, while the remaining 30% were reserved to evaluate the models' ability to predict the 'price' variable. The data was split using the KNIME Table Partitioning node. This approach allows assessing model performance on unseen data, reflecting a real situation. Models were trained using all features available after the preprocessing phase. Model performance was evaluated using quantitative metrics such as R^2 and RMSE.

3.2 Cross Validation

Models were tested using a 5-fold cross-validation methodology, using a fixed random seed to ensure reproducibility of the experiment: the dataset was divided into five subsets and each model was trained iteratively on each combination, using four subsets for training and the remaining one for testing. This approach allows assessing model performance on multiple data splits, providing a more robust and reliable evaluation. Models were trained using all features available after the preprocessing phase. Performance metrics such as R^2 and RMSE were computed for each fold and used for model comparison.

3.3 Feature Selection

Lastly, feature selection was performed to identify the most relevant features for predicting the price value. This process reduces the dimensionality of the dataset, which can speed up the training process and improve model performance. It also helps interpretability, making the predictions easier to understand. Moreover, feature selection helps prevent overfitting and allows for a clearer interpretation of the relationship between predictors and the target variable.

This approach was implemented using the Knime nodes Feature Selection Loop Start (1:1) and Feature Selection Loop End. A backward elimination strategy was applied, aiming to maximise the R2 metric: the algorithm starts with all features and removes the least important one at each iteration. Feature selection was performed using a 3-fold cross validation procedure, providing a balance between computational efficiency and the reliability of the results.

4. Evaluation

The evaluation phase aims to assess and compare the predictive performance of the machine learning models introduced in the previous section, with particular attention to their ability to generalise to unseen data.

4.1 Holdout

The performance of the models under the Holdout validation strategy was evaluated using the coefficient of determination (R^2) and the Root Mean Squared Error (RMSE), to assess both goodness of fit and predictive accuracy.

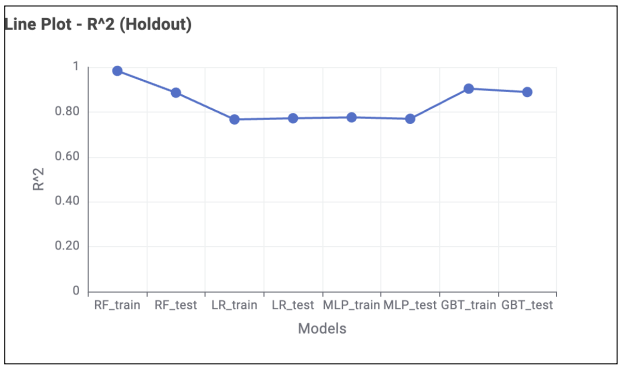


Figure 3: Line plot of R^2 comparison

As shown in Figure 3, on the training set, the model achieving the highest R^2 score is the Random Forest, with a value of 0.983, indicating an almost perfect fit to the training data. This result highlights the strong capacity of the model to capture complex relationships in the dataset. The Gradient Boosted Trees also perform well on the training set, with an R^2 of 0.903, while Linear Regression and the Multilayer Perceptron obtain significantly lower but comparable scores (0.766 and 0.775, respectively), suggesting a more limited ability to model the underlying data patterns.

Regarding the test set, the best performing model is Gradient Boosted Trees, achieving an R^2 score of 0.888, closely followed by Random Forest with 0.885. Although the models achieves a very high R^2 on the training set, which may suggest potential overfitting, the relatively small decrease in performance from training to test set indicates good generalization performance and supports the reliability of the model.

Linear Regression and the Multilayer Perceptron show very similar R^2 values on both training and test sets (from 0.766 to 0.771 and from 0.775 to 0.769, respectively), indicating stable but less accurate models.

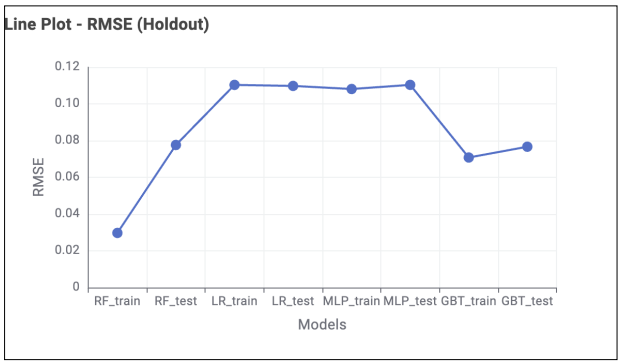


Figure 4: Line plot of RMSE comparison

The RMSE results confirm these observations, as shown in Figure 4. The Random Forest achieves the lowest error on both the training set (0.030) and the test set (0.078), while Gradient Boosted Trees also show low and consistent errors (0.071 on training and 0.077 on test). Conversely, Linear Regression and the Multilayer Perceptron exhibit higher RMSE values on both training and test sets (approximately 0.11 and 0.108, respectively), in line with their lower R^2 scores.

Overall, the evaluation highlights that ensemble methods, particularly Random Forest and Gradient Boosted Trees, provide the most accurate and robust predictions using the Holdout method. Simpler models, such as Linear Regression and the Multilayer Perceptron, demonstrate stable behaviour but inferior predictive performance. No model shows clear signs of severe overfitting, while underfitting is more evident for the simpler approaches.

4.2 Cross Validation

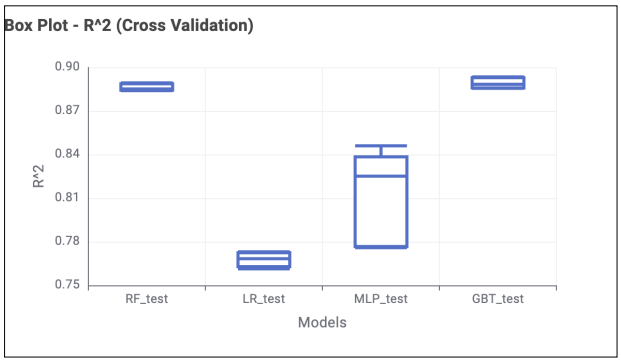


Figure 5: Box plot of R^2 comparison

The box plots provide a comparative evaluation of the models based on two performance metrics: the

coefficient of determination (R^2) and the Root Mean Squared Error (RMSE), , computed across the different folds of the cross-validation procedure.

As shown in Figure 5, which reports the distribution of R^2 values, Random Forest and Gradient Boosted Trees clearly emerge as the best-performing models. Both exhibit R^2 values consistently close to 1, indicating a strong explanatory power and a high ability to capture the variability of the target variable. Moreover, the relatively low variability across folds suggests that these models are robust and stable with respect to different train–validation splits.

In contrast, Linear Regression shows significantly lower R^2 values, while Multilayer Perceptron achieves intermediate performance, with a wider distribution that indicates less consistent behaviour across folds.

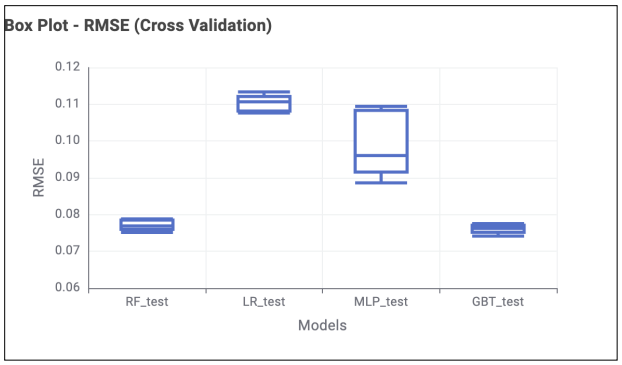


Figure 6: Box plot of RMSE comparison

The Figure 6 presents the RMSE distributions, where lower values correspond to better predictive accuracy. Consistently with the R^2-based analysis, Random Forest and Gradient Boosted Trees again achieve the lowest RMSE values, confirming their superior predictive performance. Linear Regression displays the highest RMSE, suggesting weaker generalisation capabilities, while Multilayer Perceptron shows moderate but less stable results.

Overall, the cross-validation analysis confirms that ensemble-based models not only achieve higher average performance but also exhibit greater stability across different data splits, reinforcing their suitability for this prediction task.

4.3 Feature Selection

The results of the feature selection process are illustrated in Figure 7, which shows how the R^2 score varies as a function of the number of retained features for each of the four models. This analysis highlights the impact of feature reduction on the predictive performance of each algorithm.

The Random Forest model shows a gradual improvement in performance as less informative features related to surface measures, room

configuration and structural details are removed, reaching a maximum R^2 of 0.891 with 10 features.

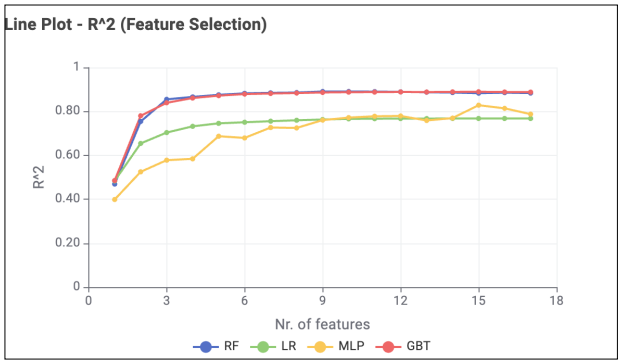


Figure 7: Line plot of R^2 comparison

Removing additional low-impact features has little effect on predictive performance, while features associated with location and property size emerge as crucial, as their exclusion leads to a noticeable decrease in accuracy.

The Linear Regression model achieves its maximum performance with an R^2 score of 0.768 using 14 features. The removal of variables related to surface measures and neighbourhood averages does not negatively affect performance, whereas excluding features associated with quality, location and property size results in a significant drop in R^2, highlighting their central role in the model.

The Multilayer Perceptron reaches its peak predictive performance with an R^2 score of 0.828 using 15 features, suggesting that most variables contribute meaningfully to the prediction task. Only features related to surface measurements and room configuration can be removed without substantially affecting performance. Conversely, the model is highly sensitive to features associated with location, quality and neighbourhood context, whose removal causes a marked decrease in predictive accuracy.

Finally, Gradient Boosted Trees show improved performance as less important features, such as those related to neighbourhood averages, room configuration and structural details, are removed, reaching a maximum R^2 of 0.889 with 14 features. The model remains stable when further minor features are eliminated, but its performance drops significantly when features associated with property size and location are excluded.

Overall, these results confirm that feature selection enables a reduction in model complexity while maintaining, and in some cases slightly improving, predictive performance. Across all models, features related to property size, location and quality consistently emerge as the most informative, whereas variables describing surface measures, room configuration and structural details are less influential.

Conclusions

The analysis demonstrates that residential property prices in King County can be effectively predicted using structural and geographic characteristics, providing a clear answer to the prediction problem posed at the outset of the study. At the same time, the results allow for the identification of the features that contribute most significantly to predictive performance, highlighting the role of both property-specific and contextual attributes.

From a methodological perspective, the preprocessing phase proved to be a crucial component of the analysis. In particular, the logarithmic transformation of the target variable addressed the strong skewness of the price distribution, while the transformation and standardisation of explanatory variables ensured compatibility across different modeling approaches, improving both numerical stability and comparability of results.

The comparison of models and training strategies consistently highlights the superior performance of ensemble-based methods, especially Random Forest and Gradient Boosted Trees, relative to simpler approaches. This outcome is well aligned with ensemble learning theory, which emphasises the ability of aggregated learners to capture nonlinear relationships and complex interactions, commonly observed in real estate markets. In contrast, parametric models such as Linear Regression, while offering greater interpretability, appear constrained by their underlying linear assumptions, resulting in lower predictive accuracy.

The evaluation of validation strategies further indicates that Cross Validation provides more stable and reliable performance estimates than the Holdout approach, while still confirming good generalisation ability for the best-performing models. Feature selection also plays an important interpretative role, reinforcing patterns already identified during the preprocessing phase, where strong multicollinearity among highly correlated variables, particularly different measures of property size, had been observed. The progressive removal of redundant features therefore led to expected results, showing that most predictive information is concentrated in a limited set of key attributes.

Overall, the results show that flexible models, combined with appropriate validation strategies and applied to a carefully preprocessed dataset, allow for accurate and stable predictions of housing prices. Nevertheless, the conclusions should be interpreted within the informational scope of the analysed dataset.

Despite the strong predictive performance, the results should be interpreted with caution. The dataset covers a specific geographical area and time period, so the learned relationships may not apply to other markets.

To address this limitation, future work could focus on evaluating model robustness and generalisation capability. This could involve testing their performance under different market conditions, for example across different geographical regions. Additionally, incorporating temporal information could help capture changes over time.

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